

Sensor Integrated Systems

Course Project Instructions

Autonomous Flame Detection and Suppression System

Purpose. This document defines the required scope, performance expectations, and submission standards for the course project.

1 Project Objective and Scope

Design and build an autonomous sensor-integrated system that detects and extinguishes a flame source located at an unknown position in the designated campus test area.

The project assesses your ability to apply measurement-system principles from the course and *Principles of Measurement Systems* (John Bentley) through integrated:

- sensing,
- signal processing,
- control,
- mechatronic actuation.

Autonomy requirement: once the start signal is given, no human intervention is allowed.

2 Team Policy

- Team size: maximum **3 students**.
- All members must contribute to design, implementation, and documentation.
- Teams must report contribution distribution at submission.
- Unless justified otherwise, contribution is assumed equal.

3 System Requirements

Each team must deliver a complete autonomous platform that can:

1. search the test area,
2. detect a flame source,
3. navigate and position for suppression,
4. extinguish the flame safely.

3.1 Mobility Platform

Wheeled locomotion is the recommended baseline. Alternative platforms (e.g., aerial or other autonomous mobility types) are acceptable when safety and task completion are fully satisfied.

If you choose a non-wheeled platform, justify key trade-offs in your report: stability, control complexity, sensing constraints, power usage, and safety.

3.2 Sensor Selection and Measurement Quality

Select and justify sensors based on relevant measurement characteristics:

- accuracy (trueness), precision/repeatability, resolution, sensitivity,
- range, noise floor, bandwidth,
- drift, linearity, cross-sensitivity.

3.3 Signal Processing and Control

- Implement filtering/processing to extract reliable measurements under noise and environmental disturbance.

- Implement control logic for search, target approach, positioning, and extinguishing activation.

3.4 Mechanical Suppression

Include a reliable extinguishing mechanism compatible with your platform and operating conditions.

4 Demonstration and Deliverables

A. Narrated Demonstration Video (mandatory)

- Start from designated start point.
- Show autonomous search and flame detection.
- Show navigation to the flame and successful extinguishing.
- Include clear narration (voice-over or on-screen) explaining architecture, sensor choices, control logic, and engineering decisions.

B. Technical Submission (mandatory)

- Technical report covering: system architecture, sensor analysis, signal processing, control strategy, and experimental results.
- Source code and system diagrams.
- Team contribution statement.

5 Evaluation Rubric

Category	Weight
System performance	30%
Sensor selection and justification	20%
Engineering design and implementation	20%
Understanding of measurement principles	15%
Quality of presentation and explanation	10%
Teamwork documentation	5%

6 Budget and Safety

- Total hardware cost must remain within the project budget specified by the instructor.
- All testing and demonstrations must follow safe laboratory practice.

Submission Note: Incomplete demonstrations or missing required artifacts may result in grade penalties.